University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

Second Midterm Exam – Version A

November 22, 2012

Surname _____ First Name _____

Student # _____ DGD (Tuesday/Thursday) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	5	3	4	3	4	5	24
Grade							

1. Let

$$A = \begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 10 & -1 & 5 \\ 2 & -5 & 4 & -3 \\ -4 & 12 & 2 & -3 \end{bmatrix}.$$

(a) [**2.5 points**] Calculate $\det A$ by using the row reduction method.

(b) **[1.5 points]** Find C_{23}, the (2,3)-cofactor of A.

(c) **[1 point]** (This part is independent of (a) and (b) above.)

Suppose A is an $n \times n$ invertible matrix. Which of the following statements is **false**? Circle the correct answer.

(a) A^{-1} is invertible.
(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
(c) If B is invertible then AB is invertible.
(d) A^T is invertible.
(e) The determinant of A is 0.

2. Determine if the following sets are subspaces of $\mathbb{R}^3$ or not. Justify your answer.
   (a) **[1.5 points]**

$$U = \left\{ \begin{bmatrix} a-b \\ a+b \\ 3a \end{bmatrix} \middle| a,\, b \in \mathbb{R} \right\}$$

(b) **[1.5 points]**

$$V = \left\{ \begin{bmatrix} a \\ b \\ ab \end{bmatrix} \middle| a,\, b \in \mathbb{R} \right\}$$

3. Consider the matrix

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 3 & 11 & 2 \end{bmatrix}.$$

(a) [**3 points**] Find the inverse of the matrix A.

(b) **[1 point]** Use A^{-1} to solve $A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

4. (a) **[1.5 points]** Solve the matrix equation

$$BX^TA^{-1} + CB = 0,$$

for X in terms of A, B, C. Here A, B, C and X are 2×2 invertible matrices.

(b) **[1.5 points]** Let A, B and C be 2×2 matrices with

$$\det A = 6, \quad \det B = -3, \quad \det\left(2A^TB^{-1}C\right) = 40.$$

Compute $\det C$.

5. Let $A = \begin{bmatrix} 1 & 0 & -2 & -4 \\ -1 & 1 & -1 & 4 \\ -2 & -2 & 10 & 8 \end{bmatrix}$. The matrix $\begin{bmatrix} 1 & 0 & -2 & -4 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is the row reduced echelon form of A.

(a) [**2 points**] Find a basis for Col A. What is rank A?

(b) [**2 points**] Find a basis for NulA.

6. An economy consists of two sectors: Agriculture and Construction. For each unit of output, Agriculture requires 0.50 units from Agriculture and 0.70 units from Construction. For each unit of output, Construction uses 0.40 units from Agriculture and 0.40 units from Construction.

(a) [**1 point**] What is the consumption matrix C for this economy?

(b) [**2 points**] Find $I - C$ and $(I - C)^{-1}$.

(c) [**1 point**] Determine what intermediate demands are created if Agriculture plans to produce 100 units.

(d) [**1 point**] Find the production levels that will satisfy the final demand of 100 units from Agriculture and 200 units from Construction.

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